PROJECT PROGRESS REPORT

BLE-Based Smart Workforce Presence and Availability Monitoring System

Using ESP32 and IoT Analytics

TEAM	Program
Group No. 1	IoT , Robotics and Drones
GUIDED BY	DATE
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PROJECT CONTEXT

Problem Statement

- Organizations require real-time information about employee availability and room occupancy.
- Existing solutions based on CCTV, RFID, or GPS are expensive, power hungry, or unsuitable for indoor environments.
- There is a need for a low-cost, scalable indoor presence monitoring system.
- Traditional attendance systems only capture entry and exit, not current availability.

INDOOR PRESENCE — TECHNOLOGY COMPARISON

RFID

Short range, tag cost

GPS

No indoor accuracy

CCTV

High cost, privacy concerns

BLE + ESP32

Low cost, scalable, indoor-ready

PROJECT SCOPE

Project Objectives

PRIMARY OBJECTIVE

Develop a low-cost indoor workforce presence monitoring system using BLE-enabled ESP32 devices.

SECONDARY OBJECTIVES

- Detect employee presence inside different zones or rooms.
- Provide real-time occupancy information.
- Estimate proximity using BLE RSSI values.
- Enable future dashboard analytics.
- Communicate data through IoT infrastructure.

SYSTEM DESIGN

Hardware Selection and System Components

COMPONENTS SELECTED SO FAR

- ESP32 development boards for badges and scanners.
- BLE communication using onboard Bluetooth capability.
- Buzzer for alerts and notifications.
- Power adapter or battery-powered badge operation.
- Optional future cloud dashboard and MQTT broker.

Why ESP32?

Selected for integrated WiFi and BLE support, low cost, and an easy programming environment.

SYSTEM HARDWARE BLOCK OVERVIEW

ESP32 Badge/Scanner	BLE Radio	Buzzer Alerts
Power / Battery	MQTT Broker (future)	Dashboard (future)

EXPERIMENTAL WORK

BLE Communication Testing

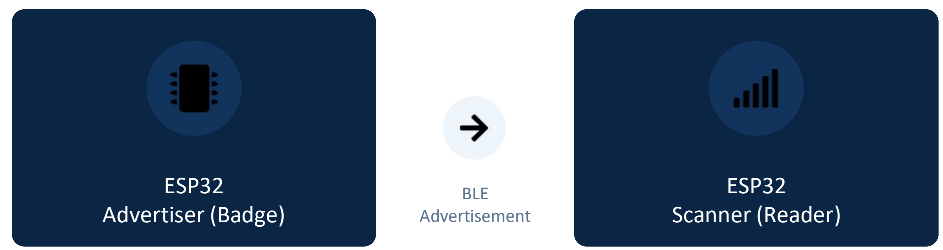
WORK COMPLETED

- Configured ESP32 as BLE advertiser.
- Configured ESP32 as BLE scanner.
- Successfully detected nearby devices.
- Verified advertisement packet reception.
- Measured RSSI values at different distances.

OBSERVATIONS

Communication worked reliably within room-scale distances. RSSI values varied based on distance and environment.

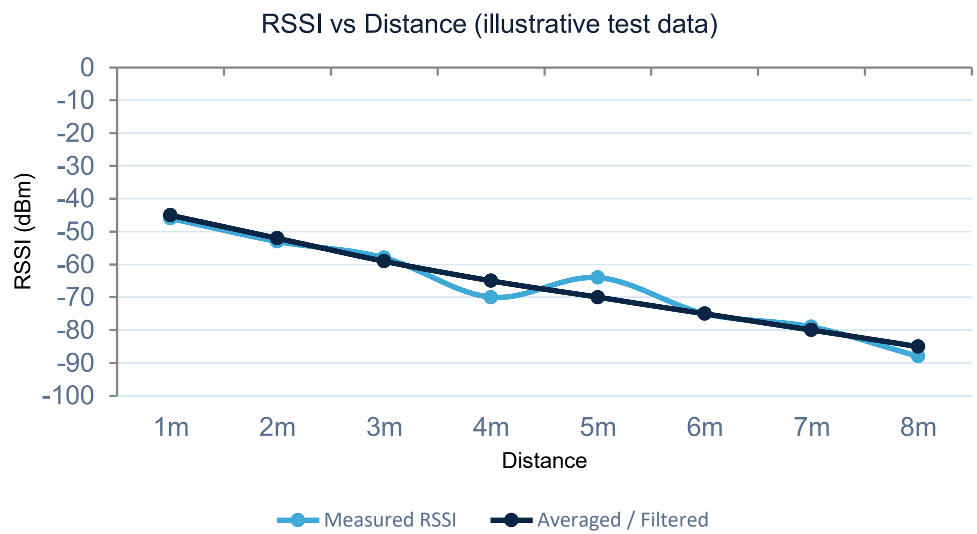
BLE COMMUNICATION FLOW



RSSI signal strength decreases as distance between badge and scanner increases.

DATA ANALYSIS

RSSI Analysis and Distance Estimation Challenges



RSSI fluctuates even at fixed positions — human bodies, walls, and reflections affect measurements, and the RSSI–distance relationship is highly non-linear.

EXPERIMENTS PERFORMED

- Recorded RSSI values at various distances.
- Applied median filtering and averaging techniques.
- Studied RSSI behaviour inside rooms.

CONCLUSION

Exact positioning is difficult using BLE RSSI alone.

Signal reflections from walls, furniture, and human bodies introduce noise that pure trilateration cannot reliably filter out at room scale.

IOT COMMUNICATION

MQTT-Based IoT Architecture

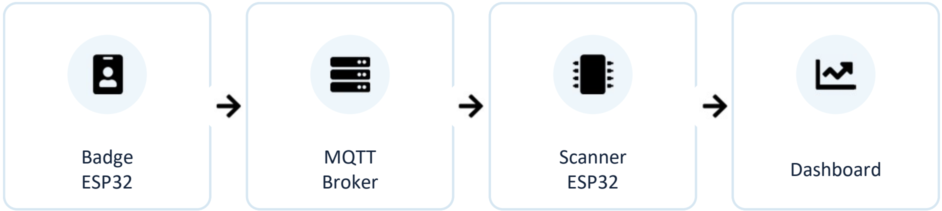
PROGRESS ACHIEVED

- Explored MQTT communication between ESP32 devices.
- Studied broker-based communication architecture.
- Understood publisher and subscriber model.
- Tested data transmission concepts.

WHY MQTT?

MQTT provides lightweight, low-bandwidth communication well suited to constrained IoT devices such as the ESP32, using a publish/subscribe model over a central broker.

ARCHITECTURE FLOW



Badge ESP32 → MQTT Broker → Scanner ESP32 → Dashboard

Publisher / Subscriber model decouples badges from the dashboard — devices communicate only through the broker, simplifying scaling to many nodes.

HARDWARE ENCLOSURE

Wearable Badge Design Progress

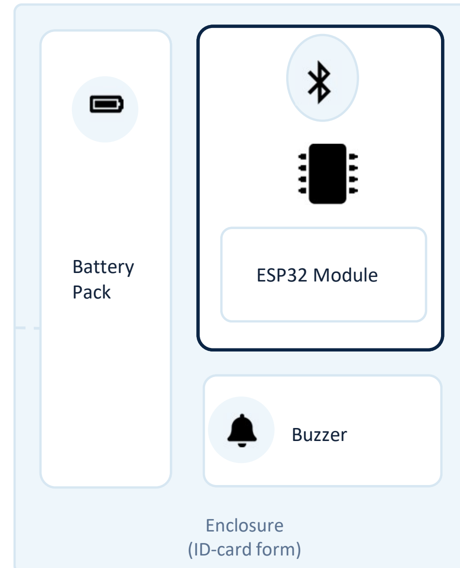
DESIGN ACTIVITIES COMPLETED

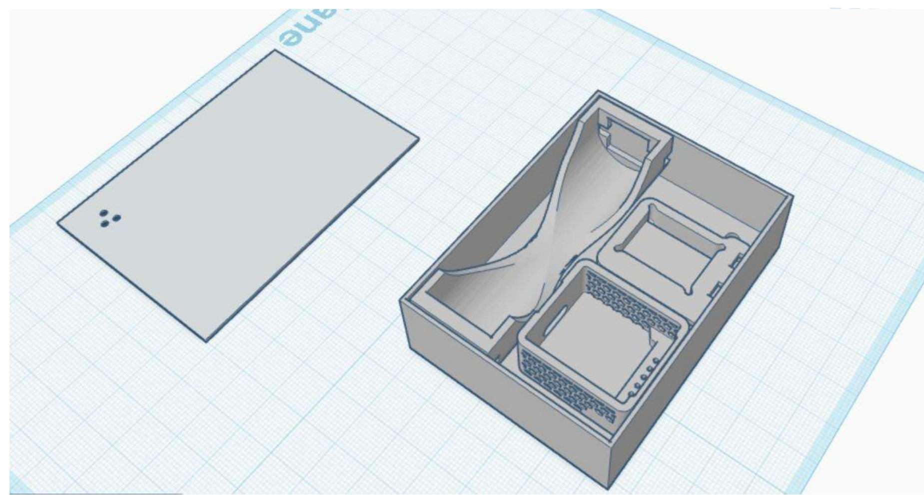
- Started development of an ID-card style enclosure.
- Created cavity dimensions for ESP32 placement.
- Designed space for buzzer integration.
- Considered battery placement and accessibility.

DESIGN GOALS

- Comfortable wearable form factor
- Professional appearance
- Easy charging and maintenance

BADGE — EXPLODED LAYOUT VIEW





LEARNINGS

Challenges and Learnings

MAJOR CHALLENGES ENCOUNTERED

- RSSI instability
- Multipath signal reflections
- Non-linear distance estimation
- Environmental interference
- Difficulty in precise indoor positioning

KEY LEARNING

Instead of exact coordinates, room-level or zone-level localization is more practical and reliable.

This shift in approach — from precise trilateration to zone-based presence detection — better matches what BLE hardware can reliably deliver in real indoor environments.

PROJECT STATUS

Current Progress and Future Work

CURRENT STATUS

- BLE communication validated
- ESP32 hardware selected and tested
- RSSI experiments completed
- MQTT architecture studied
- Badge enclosure design initiated

NEXT STEPS

- Connect multiple scanners simultaneously
- Implement zone-based detection
- Develop backend data collection system
- Create visualization dashboard
- Perform real-world testing in multiple rooms

PROOF-OF-CONCEPT COMPLETION

70%

The project has successfully completed the proof-of-concept stage and is moving toward multi-node deployment and analytics integration.